Designing size-aware cryptographic primitives for FPGA-based accelerators

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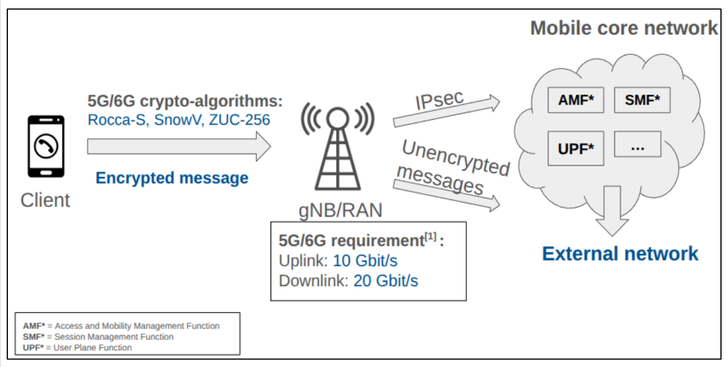
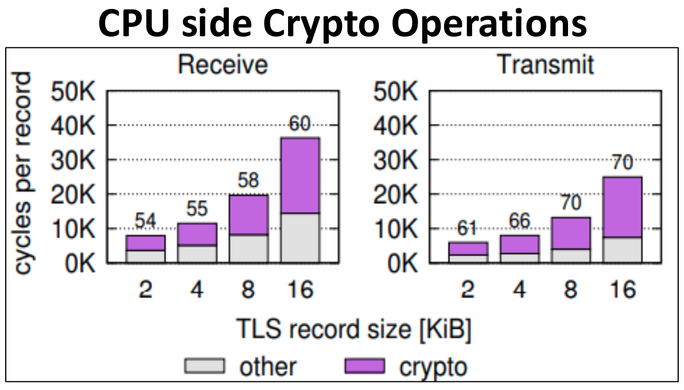
Background

Cryptographic primitives use at Radio Access Network



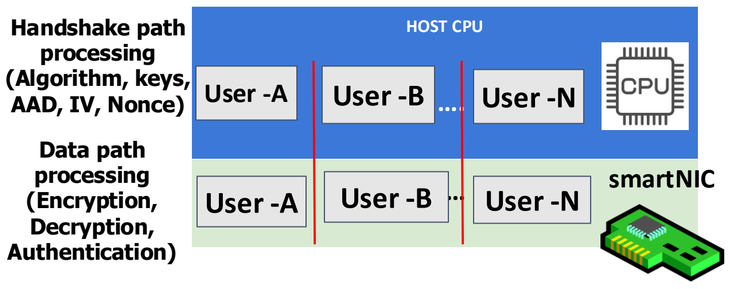
Other Applications:
- Telecom Networks , Military Applications, Data Centers

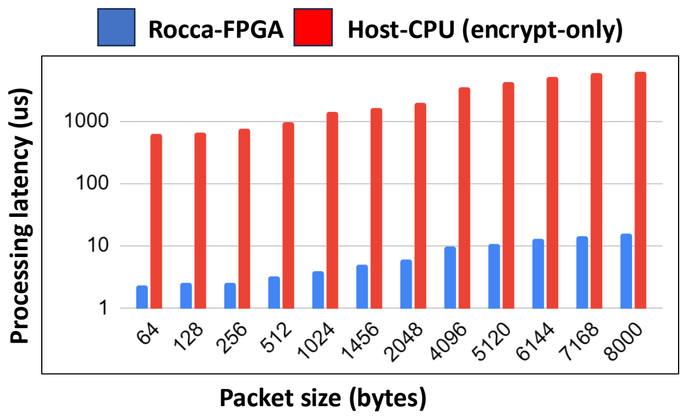
CPU side Crypto Operations



Encryption/decryption[1]
- Simple TCP bandwidth test
- Modified Iperf to use OppenSSL
- 60-74% CPU cycles spent for TLS TX/RX

Server accelerates Data Centers applications by leveraging Smart NICs





Rocca-S processing latency
❖Offload < Host processing
➢Up to 99.75%

State-of-the-art solutions

Fixed function ASIC solutions Not flexible	Reconfigurable solutions Flexible
• Nvidia's Bluefield[2] • AMD Pensando's DSC • Netronome CX NICs	• FPGA-based solutions • Chacha on programmable NIC[3] • AES on programmable Tofino switch

Limitations of the State-of-the-art

- CPUs are general-purpose, may lead to performance bottlenecks
  - CPU Specialized Accelerators (AES-NI)
    - Consumes lot of CPU cycles !
- ASICs are fast but fixed function
- Programmable smart NICs
  - Reconfigurable but have limited resources
  - Focus on individual algorithm

The above solutions are optimized to be resource , power, throughput efficient but are not workload aware !

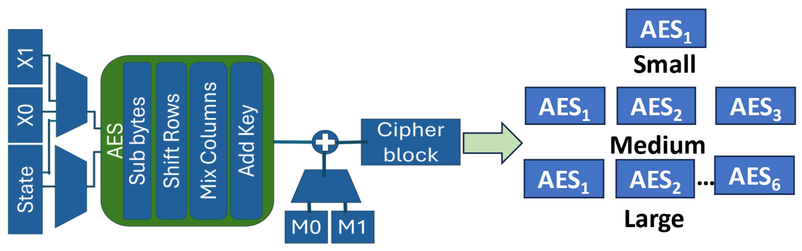
Resource, power or throughput trade-off for FPGAs ?

- As 5G telecom algorithms continue to evolve, there is a need for:
  - Programmability
  - Configurability

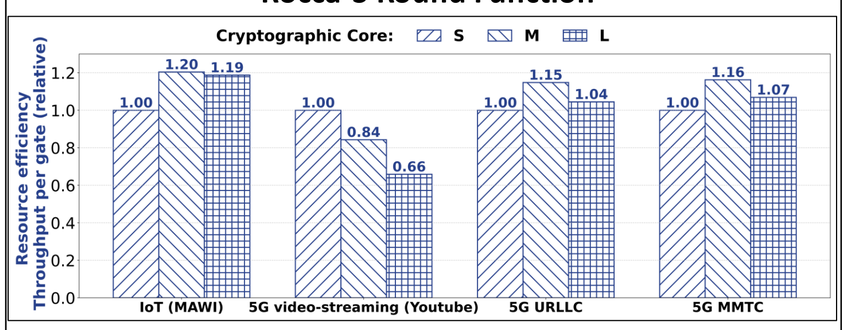
Earlier FPGA solutions focused on either:
- High throughput => Using more resources
- Resource efficient(Reusing the components) =>Increase in latency (sequential processing)

Our Problem Statement & Design

Cryptographic Core Design = {Small, Medium$_n$, Large}



Rocca-S Round Function



Rocca-S's resource efficiency across cryptographic core variants for 5G applications relative to the cryptographic core, "S".

- ➢M Cryptographic core => Resource efficient ↑ for 5G datasets
- ➢S Cryptographic core => Resource efficient ↑ for 5G video streaming dataset
- ➢L Cryptographic core ≅ M cryptographic core for IOT dataset

Design cryptographic primitive offload framework that to :
- Overcome trade-off between performance & resource utilization
- Reconfigurable
  - Dynamically reconfigurable (DPR)
- Workload-aware
- Asymmetric Replication → Workload-aware

Scheduler Design

Scheduler Toy example

Arrival time	0	0	0	2	0	1	2	2	3
Request ID (rij)	r_{11}	r_{12}	r_{13}	r_{14}	r_{21}	r_{22}	r_{23}	r_{24}	r_{25}
Service time	1	1	2	2	1	1	1	2	1
Request len (ri.len)	300	500	2000	8000	100	400	200	4000	500
Rank (ri.rank)	4	3	2	0	4	4	4	1	3

Requests arrived at the NIC; rij= jth request of flow i

FlexMesh$_{nWC}$

Virtual time		1	2	3	4	5	6	7
Cryptographic core (C_1)		r_{11}	r_{21}	r_{12}	r_{14}		r_{23}	r_{25}
Cryptographic core (C_2)		r_{13}		r_{22}	r_{24}			
Wire Output		r_{11}	r_{21}	r_{13}	r_{12}	r_{22}	r_{14}	r_{24}	r_{23}	r_{25}

Non-work conserving (2 idle) , OOO Size=14.5 KB, total time = 7

FlexMesh

Virtual time		1	2	3	4	5	6
Cryptographic core (C_1)	HEAD	r_{11}	r_{21}	r_{12}	r_{14}		r_{23}
Cryptographic core (C_2)		r_{13}		r_{22}	r_{24}		r_{25}
Wire Output	HEAD	r_{11}	r_{21}	r_{13}	r_{12}	r_{22}	r_{14}	r_{24}	r_{23}	r_{25}

Work-conserving, OOO Size = 14.5 KB, total time = 6

FlexMesh$_p$

Virtual time		1	2	3	4	5	6
Eligible requests (E1)	HEAD	---	r_{21}, r_{12}	r_{21}, r_{14}, r_{23}		r_{21}, r_{23}, r_{25}	r_{21}, r_{23}
Rank ($rank_1$)		---	4, 3	4, 0, 4		4, 4, 3	4, 4
Cryptographic core (C_1)		r_{11}	r_{12}	r_{14}		r_{25}	r_{21}
Eligible requests (E_2)		---		r_{22}, r_{24}		r_{22}	
Rank ($rank_2$)				4, 1		4	
Cryptographic core (C_2)		r_{13}		r_{24}		r_{22}	r_{23}
Wire output	HEAD	r_{11}	r_{12}	r_{13}	r_{14}	r_{24}	r_{22}	r_{25}	r_{21}	r_{23}

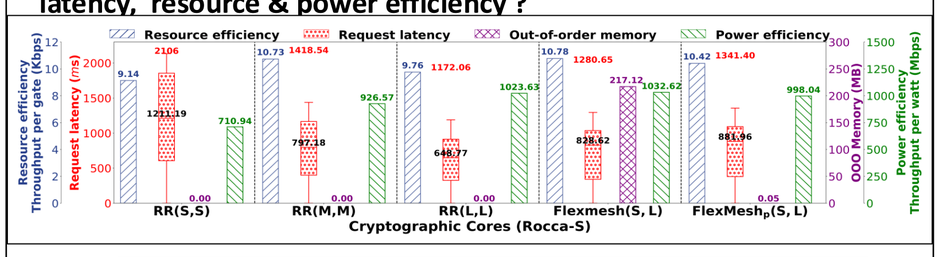
Work-conserving, OOO size = 4.9 KB, total time = 6

rij – jth request of flow i, Out-of-order, Work-conserving, Idle

Experimental Setup & Results

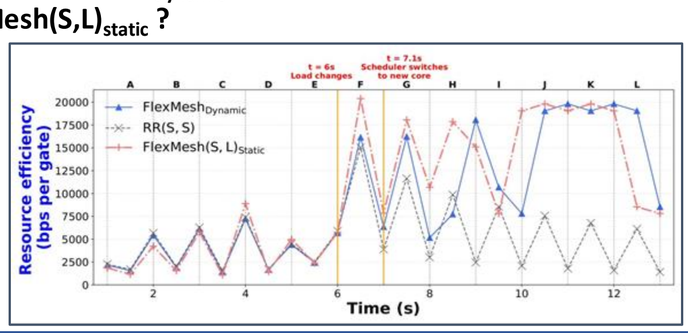
- ➢Zynq UltraScale+ MPSoC ZCU106)
  - Rocca-S implementation on (ZCU-106+ board)

➢ **How different scheduling policies impact OOO memory, request latency, resource & power efficiency ?**



FlexMesh => Resource ↑, Power ↑ & Lower request Latency ↑

➢ **How do FlexMesh$_{dynamic}$ resource efficient w.r.t to RR(S,S) & FlexMesh(S,L)$_{static}$?**



- ➢FlexMesh$_{dynamic}$ ≃ RR(S,S) > FlexMesh(S,L)$_{static}$ (t < 6 seconds)
- ➢FlexMesh$_{dynamic}$ ≃ FlexMesh(S,L) > RR(S,S) (t > 7.1 seconds)

Conclusion

➢We present FlexMesh, an in-network cryptographic primitive that leverages the workload request size characteristics to design cryptographic accelerators with high resource and power efficiency, without compromising the performance.

➢We design and implement variable-sized cryptographic primitives for next-generation mobile networks (i.e., Rocca-S), and a programmable scheduler that prioritizes flow order and work-conservation.

References

[1] Boris Pismenny, Haggai Eran, Aviad Yehezkel, Liran Liss, Adam Morrison, and Dan Tsafrir. 2021. Autonomous NIC Offloads
[2] 2021. NVIDIA BLUEFIELD-3 DPU PROGRAMMABLE DATA CENTER INFRASTRUCTURE ON-A-CHIP. https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/documents/datasheet-nvidia-bluefield-3-dpu.pdf.
[3]Shaguftha Zuveria Kottur, Krishna Kadiyala, Praveen Tammana, and Rinku Shah. 2022. Implementing ChaCha Based Crypto Primitives on Programmable SmartNICs.

Have a Chai with the authors

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Paper link

GitHub code available